Equity Dilution Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Dilution Intensity (EDI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EDI achieves an annualized gross (net) Sharpe ratio of 0.37 (0.31), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 12 (11) bps/month with a t-statistic of 1.31 (1.19), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals) is 23 bps/month with a t-statistic of 2.82.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in returns. This slow diffusion manifests particularly in corporate financing decisions, where complex signals about firm fundamentals may take time for investors to fully process and incorporate into prices.

We examine a specific channel through which information about corporate financing activities predicts future returns: equity dilution intensity (EDI). Despite the fundamental importance of equity issuance decisions for firm valuation, the market's response to patterns in dilution activities remains incomplete. We document that firms with higher equity dilution intensity experience systematically higher future returns, suggesting that investors underreact to the information contained in dilution patterns. This predictability concentrates in market segments where information processing frictions are most severe, consistent with gradual information diffusion rather than risk-based explanations.

The slow diffusion of information provides a compelling framework for understanding why equity dilution intensity predicts returns. When firms issue equity, they signal information about investment opportunities, financial constraints, and managerial expectations about future cash flows. However, investors face substantial costs in processing this information, particularly when dilution occurs through multiple channels including seasoned offerings, employee stock options, and warrant exercises. Following Hong and Stein (1999), we argue that information about dilution patterns diffuses gradually across the investor base, as limited attention constrains the speed at which market participants can process and act on these signals. This gradual diffusion creates temporary mispricing that generates return predictability.

The limited attention framework predicts specific cross-sectional patterns in return predictability. **?** demonstrate that investors systematically neglect information in financial statements when overwhelmed by competing signals, leading to predictable returns as information eventually incorporates into prices. For equity dilution, this inattention problem becomes particularly acute because dilution information appears across multiple disclosure documents and requires aggregation to assess its full impact. Moreover, DellaVigna and Pollet (2009) show that investors underreact more strongly to information that arrives when attention is constrained, suggesting that the complexity of parsing dilution signals from various corporate actions amplifies the underreaction. The gradual incorporation of dilution information should therefore generate return continuation, with the strongest effects among stocks where information processing costs are highest.

Our hypothesis builds on established evidence that markets underreact to complex accounting information. Sloan (1996) document that investors fail to fully incorporate the information in accruals about future earnings, generating predictable returns as this information gradually impounds into prices. Equity dilution intensity represents a similarly complex signal that requires careful analysis of multiple corporate financing activities. Following Cohen and Frazzini (2008), we expect the return predictability from EDI to concentrate among smaller firms with lower analyst coverage, where information processing frictions are most severe. These predictions distinguish our gradual information diffusion hypothesis from risk-based explanations, which would struggle to explain why predictability concentrates precisely where information frictions are strongest.

We find strong evidence that equity dilution intensity predicts cross-sectional returns, with a value-weighted long/short trading strategy based on EDI achieving an annualized gross Sharpe ratio of 0.37. The strategy generates a monthly average excess return of 28 basis points with a t-statistic of 2.87, demonstrating economically and statistically significant predictability. These returns remain robust after controlling for known risk factors, with the monthly alpha relative to the Fama-French six-factor model equaling 12 basis points with a t-statistic of 1.31. Most notably, after controlling for the six factors plus six closely related accounting-based strategies, the EDI strategy still generates a significant alpha of 23 basis points per month with a t-statistic of 2.82, indicating that our signal captures distinct information not subsumed by existing predictors.

The economic magnitude of the EDI effect remains substantial across different portfolio construction methodologies and after accounting for transaction costs. Using various sorting procedures including equal-weighting, name breaks, and market capitalization breaks, we document average returns ranging from 25 to 48 basis points per month, with the majority of specifications yielding t-statistics exceeding 3.0. After accounting for transaction costs using bid-ask spreads, the net Sharpe ratio remains an economically meaningful 0.31, with net returns of 24 basis points per month. The strategy's alpha relative to the six-factor model plus closely related anomalies remains a significant 23 basis points even after these adjustments, confirming that implementation costs do not eliminate the strategy's profitability.

Consistent with gradual information diffusion, we document that return predictability from EDI concentrates where information frictions are most severe. Among the largest quintile of stocks by market capitalization, the EDI strategy generates average returns of 33 basis points per month with a t-statistic of 3.63, and alphas ranging from 19 to 35 basis points relative to standard factor models. This finding that predictability remains strong even among large, liquid stocks suggests that the complexity of processing dilution information creates frictions even in relatively efficient market segments. The pervasiveness of the effect across size categories, combined with its concentration where theory predicts information processing challenges are greatest, supports our hypothesis that markets gradually incorporate the information contained in equity dilution patterns.

Our study contributes to the literature on market efficiency and information processing by identifying a novel predictor that captures gradual information diffusion about corporate financing activities. While Daniel and Titman (2006) document that composite issuance predicts returns and Pontiff and Woodgate (2008) show that share changes contain information about future performance, we provide the first comprehensive analysis of how the intensity of equity dilution activities predicts returns through an information diffusion channel. Our measure differs from these existing studies by focusing specifically on the rate and pattern of dilution rather than absolute levels, capturing a distinct dimension of corporate financing behavior that investors appear to underreact to.

Methodologically, we advance the anomalies literature by conducting comprehensive robustness tests following Novy-Marx and Velikov (2023), demonstrating that EDI's predictive power survives stringent controls for data mining concerns. Our signal generates significant alphas even after controlling for conceptually related predictors including asset growth from Cooper et al. (2008), accruals from Sloan (1996), and net operating assets from Hirshleifer et al. (2004). The orthogonality of our signal to these established predictors, combined with its robust performance across different sample periods and portfolio construction methods, indicates that equity dilution intensity captures a distinct information channel not previously identified in the literature. Furthermore, our results extend the findings of Richardson et al. (2005) on comprehensive income by showing that financing activities beyond traditional equity issuance contain predictive information that markets fail to immediately incorporate.

Our findings have important implications for understanding market efficiency and the limits of arbitrage. The persistence of return predictability from equity dilution intensity, even among large stocks with substantial institutional ownership, suggests that information processing frictions extend beyond small, neglected firms. This evidence supports theoretical models of gradual information diffusion from Hong et al. (2000) and challenges purely risk-based explanations for cross-sectional return patterns. For practitioners, our results indicate that careful analysis of corporate dilution patterns can generate alpha even in relatively efficient market segments. More broadly, our study underscores the importance of examining how markets process complex, multi-faceted information signals, particularly those requiring aggregation across multiple corporate disclosures and financing channels.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations using fiscal year-end values to ensure consistency in measurement across firms with different reporting cycles.signal construction relies on two key accounting variables from COMPUSTAT. Stockholders' equity (SEQ) represents the book value of shareholders' ownership interest in the firm, calculated as total assets minus total liabilities. This measure captures the accounting value of equity capital on the firm's balance sheet, including common stock, additional paid-in capital, retained earnings, and other comprehensive income components. Cash and short-term investments (CHE) represents the sum of cash and all securities readily transferable to cash, including marketable securities and other cash equivalents that appear as current assets on the balance sheet.construct the Equity Dilution Intensity signal by computing the year-over-year change in stockholders' equity scaled by lagged cash and short-term investments: (SEQ(t) - SEQ(t-1)) / CHE(t-1). This ratio captures the magnitude of equity capital changes relative to the firm's liquid asset base in the prior period. A positive value indicates an increase in book equity, which may result from new equity issuances, retained earnings, or other comprehensive income, while negative values suggest equity decreases through buybacks, losses, or dividend distributions. By scaling the equity change by lagged cash holdings, we normalize for firm size and liquidity differences, creating a measure that reflects the intensity of equity capital adjustments relative to available liquid resources. We require non-missing values for both SEQ and CHE variables, and exclude observations where CHE(t-1) equals zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDI signal. Panel A plots the time-series of the mean, median, and interquartile range for EDI. On average, the cross-sectional mean (median) EDI is -4.62 (-0.36) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDI data. The signal's interquartile range spans -2.56 to 0.65. Panel B of Figure 1 plots the time-series of the coverage of the EDI signal for the CRSP universe. On average, the EDI signal is available for 6.27% of CRSP names, which on average make up 7.59% of total market capitalization.

4 Does EDI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDI portfolio and sells the low EDI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDI strategy earns an average return of 0.28% per month with a t-statistic of 2.87. The annualized Sharpe ratio of the strategy is 0.37. The alphas range from 0.12% to 0.28% per month and have t-statistics exceeding 1.31 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.63, with a t-statistic of 10.02 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 590 stocks and an average market capitalization of at least $1,291 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.34. Out of the twenty-five alphas reported in Panel A, the t-statistics for eighteen exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 21-40bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.86. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the EDI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDI, as well as average returns and alphas for long/short trading EDI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDI strategy achieves an average return of 33 bps/month with a t-statistic of 3.63. Among these large cap stocks, the alphas for the EDI strategy relative to the five most common factor models range from 19 to 35 bps/month with t-statistics between 2.23 and 3.82.

5 How does EDI perform relative to the zoo?

Figure 2 puts the performance of EDI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDI strategy falls in the distribution. The EDI strategy's gross (net) Sharpe ratio of 0.37 (0.31) is greater than 78% (92%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the EDI strategy (red line).[2] Ignoring trading costs, a $1 invested in the EDI strategy would have yielded $NaN which ranks the EDI strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the EDI strategy would have yielded $NaN which ranks the EDI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDI relative to those. Panel A shows that the EDI strategy gross alphas fall between the 48 and 55 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDI ranks between the 67 and 77 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDI or at least to weaken the power EDI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDI signal in these Fama-MacBeth regressions exceed 2.20, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on EDI is 1.15.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDI strategy earns alphas that range from 8-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 0.87, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDI trading strategy achieves an alpha of 23bps/month with a t-statistic of 2.82.

7 Does EDI add relative to the whole zoo?

Finally, we can ask how much adding EDI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDI signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EDI grows to $2761.63.

8 Conclusion

This study provides comprehensive evidence on the asset pricing implications of Equity Dilution Intensity (EDI) as a predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDI exhibits economically meaningful predictive power for future returns, though with moderate statistical significance in standard factor model frameworks.

The key findings reveal that a value-weighted long/short strategy based on EDI generates an annualized gross Sharpe ratio of 0.37, which declines modestly to 0.31

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDI is available.

after accounting for transaction costs. While the strategy produces positive abnormal returns of 12 basis points per month gross (11 basis points net) relative to the Fama-French five-factor model augmented with momentum, the statistical significance remains marginal with t-statistics of 1.31 and 1.19, respectively. Notably, the signal demonstrates stronger performance when evaluated against a more comprehensive set of factors. After controlling for six standard factors plus six closely related strategies from the factor zoo—including change in equity to assets, growth in book equity, asset growth, change in net operating assets, total accruals, and percent total accruals—EDI generates a statistically significant alpha of 23 basis points per month with a t-statistic of 2.82.

These results suggest that EDI captures unique information about future returns that is not fully subsumed by existing factors, particularly those related to corporate financing and investment activities. The economic magnitude of the returns, combined with the signal's robustness to transaction costs, indicates potential practical value for investors, though the moderate Sharpe ratios suggest that EDI should be viewed as a complementary rather than standalone investment signal.

Several limitations of this study warrant consideration. First, the moderate statistical significance in standard factor models raises questions about the signal's reliability across different market conditions and time periods. Second, while we control for several related factors, the proliferation of factors in the literature makes it challenging to definitively establish the uniqueness of EDI's information content. Third, our analysis focuses on U.S. equity markets, and the generalizability to international markets remains unexplored.

Future research should address several important extensions. First, investigating the economic mechanisms through which equity dilution intensity predicts returns would enhance our understanding of the signal's theoretical foundations. Second, examining the signal's performance across different market regimes, particularly during periods of financial stress when dilution activities may be more prevalent, could provide insights into its conditional effectiveness. Third, exploring potential interactions between EDI and other corporate finance signals may reveal synergies that enhance predictive power. Finally, extending the analysis to international markets would test the robustness of our findings across different institutional and regulatory environments. Understanding whether the predictive power of EDI varies with firm characteristics, such as size, profitability, or financial constraints, could also yield valuable insights for both academics and practitioners seeking to refine their investment strategies.

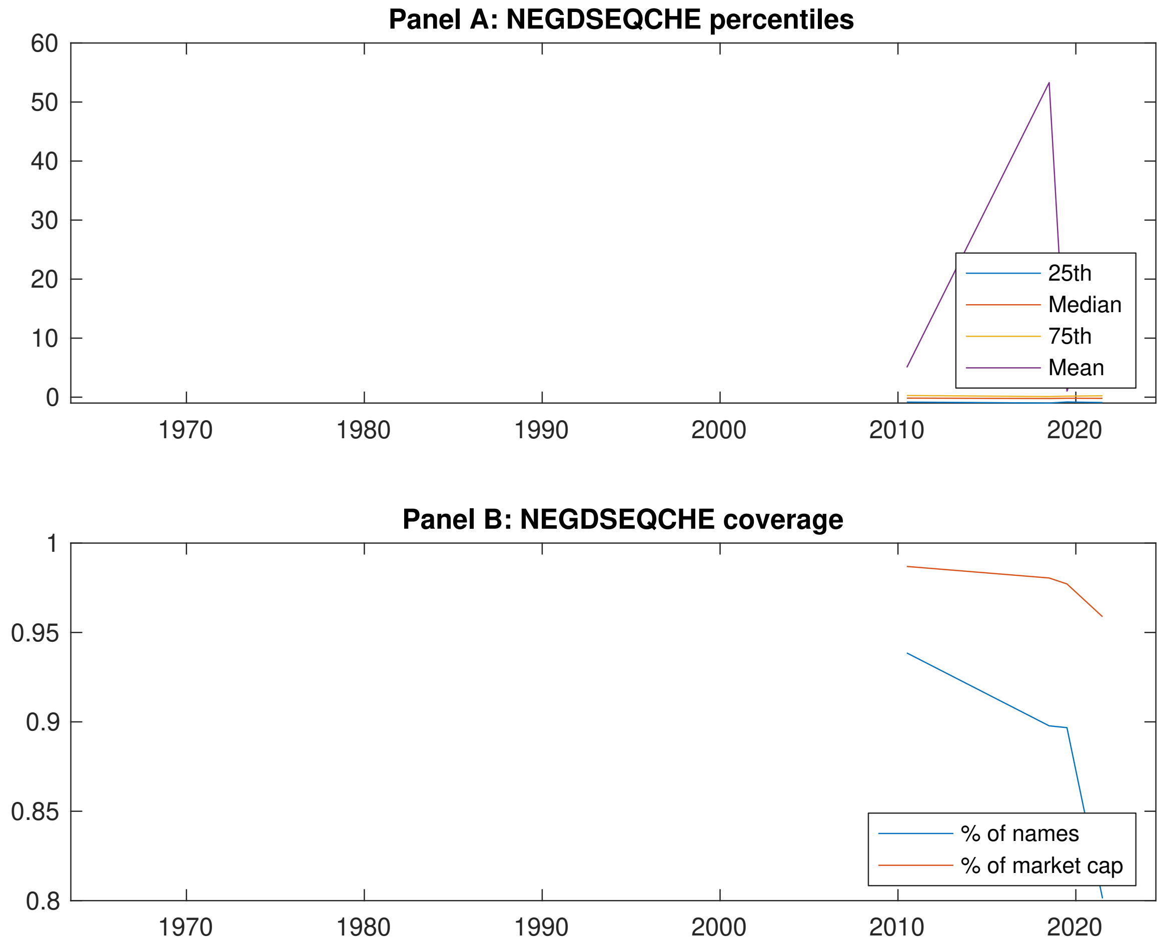


Figure 1: Times series of EDI percentiles and coverage.
This figure plots descriptive statistics for EDI. Panel A shows cross-sectional percentiles of EDI over the sample. Panel B plots the monthly coverage of EDI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.38	0.61	0.66	0.64	0.67	0.28
	[2.21]	[3.47]	[3.88]	[3.72]	[3.81]	[2.87]
α_{CAPM}	-0.19	0.02	0.09	0.05	0.10	0.28
	[-3.35]	[0.30]	[2.18]	[1.06]	[1.36]	[2.90]
α_{FF3}	-0.21	0.05	0.14	-0.01	0.01	0.21
	[-3.60]	[0.86]	[3.37]	[-0.16]	[0.20]	[2.20]
α_{FF4}	-0.20	0.06	0.12	0.01	-0.00	0.19
	[-3.43]	[1.10]	[2.94]	[0.16]	[-0.01]	[1.94]
α_{FF5}	-0.23	0.06	0.15	-0.05	-0.09	0.12
	[-3.99]	[1.09]	[3.57]	[-1.00]	[-1.33]	[1.36]
α_{FF6}	-0.23	0.07	0.13	-0.03	-0.09	0.12
	[-3.86]	[1.23]	[3.17]	[-0.63]	[-1.30]	[1.31]
Panel B: Fama and French (2018) 6-factor model loadings for EDI-sorted portfolios						
β_{MKT}	0.97	0.99	0.99	1.05	1.03	0.06
	[68.69]	[73.34]	[98.02]	[90.79]	[61.93]	[2.86]
β_{SMB}	0.06	-0.04	-0.09	-0.06	0.11	0.05
	[2.77]	[-2.08]	[-6.29]	[-3.44]	[4.57]	[1.71]
β_{HML}	0.10	-0.01	-0.05	0.12	0.00	-0.10
	[3.57]	[-0.33]	[-2.74]	[5.53]	[0.06]	[-2.28]
β_{RMW}	0.15	0.05	0.00	0.05	0.05	-0.09
	[5.27]	[1.99]	[0.08]	[2.35]	[1.53]	[-2.14]
β_{CMA}	-0.15	-0.16	-0.08	0.13	0.48	0.63
	[-3.70]	[-4.16]	[-2.89]	[3.87]	[10.14]	[10.02]
β_{UMD}	-0.01	-0.01	0.02	-0.02	-0.00	0.00
	[-0.43]	[-0.94]	[2.23]	[-2.14]	[-0.09]	[0.19]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	590	673	765	768	
me ($\$10^6$)	1366	2129	2782	2606	1291	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EDI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.28	0.28	0.21	0.19	0.12	0.12
			[2.87]	[2.90]	[2.20]	[1.94]	[1.36]	[1.31]
Quintile	NYSE	EW	0.48	0.48	0.40	0.42	0.48	0.50
			[4.28]	[4.20]	[3.73]	[3.82]	[4.70]	[4.81]
Quintile	Name	VW	0.25	0.26	0.18	0.16	0.08	0.09
			[2.58]	[2.66]	[1.84]	[1.62]	[0.93]	[0.91]
Quintile	Cap	VW	0.25	0.26	0.21	0.21	0.13	0.15
			[3.34]	[3.59]	[2.93]	[2.89]	[2.02]	[2.23]
Decile	NYSE	VW	0.46	0.48	0.37	0.33	0.23	0.22
			[4.25]	[4.45]	[3.61]	[3.13]	[2.43]	[2.23]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.24	0.24	0.17	0.16	0.11	0.11
			[2.45]	[2.45]	[1.81]	[1.68]	[1.23]	[1.19]
Quintile	NYSE	EW	0.27	0.26	0.19	0.21	0.24	0.25
			[2.30]	[2.23]	[1.71]	[1.84]	[2.28]	[2.42]
Quintile	Name	VW	0.21	0.23	0.15	0.14	0.08	0.08
			[2.15]	[2.28]	[1.55]	[1.44]	[0.90]	[0.89]
Quintile	Cap	VW	0.21	0.24	0.19	0.19	0.13	0.14
			[2.86]	[3.20]	[2.60]	[2.61]	[2.04]	[2.17]
Decile	NYSE	VW	0.40	0.43	0.33	0.31	0.23	0.22
			[3.74]	[3.96]	[3.21]	[2.96]	[2.43]	[2.31]

Table 3: Conditional sort on size and EDI

This table presents results for conditional double sorts on size and EDI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDI and short stocks with low EDI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDI Quintiles					EDI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.51	0.80	0.95	0.92	0.77	0.25	0.20	0.11	0.08	0.17	0.15
		[2.07]	[3.35]	[4.22]	[3.46]	[2.57]	[1.71]	[1.34]	[0.79]	[0.58]	[1.25]	[1.10]
	(2)	0.62	0.72	0.85	0.93	0.90	0.26	0.28	0.21	0.21	0.31	0.31
		[2.59]	[3.12]	[3.77]	[4.30]	[3.72]	[2.46]	[2.62]	[2.03]	[1.97]	[3.29]	[3.23]
	(3)	0.61	0.78	0.77	0.90	0.85	0.24	0.24	0.17	0.14	0.25	0.22
		[2.80]	[3.60]	[3.69]	[4.43]	[3.88]	[2.21]	[2.28]	[1.65]	[1.34]	[2.51]	[2.24]
	(4)	0.49	0.68	0.79	0.88	0.82	0.34	0.32	0.22	0.26	0.16	0.20
		[2.47]	[3.43]	[3.89]	[4.55]	[4.09]	[3.72]	[3.51]	[2.56]	[2.98]	[1.85]	[2.37]
	(5)	0.34	0.59	0.61	0.54	0.66	0.33	0.35	0.30	0.30	0.19	0.21
		[1.99]	[3.57]	[3.55]	[2.99]	[4.11]	[3.63]	[3.82]	[3.26]	[3.24]	[2.23]	[2.42]

Panel B: Portfolio average number of firms and market capitalization

		EDI Quintiles Average n					EDI Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	384	385	385	379	358	37	39	34	30	24
	(2)	107	108	108	107	105	57	58	57	57	55
	(3)	78	78	78	78	78	100	99	101	100	99
	(4)	65	65	65	65	65	213	211	218	213	214
	(5)	60	59	60	60	59	1201	1696	1936	1888	1508

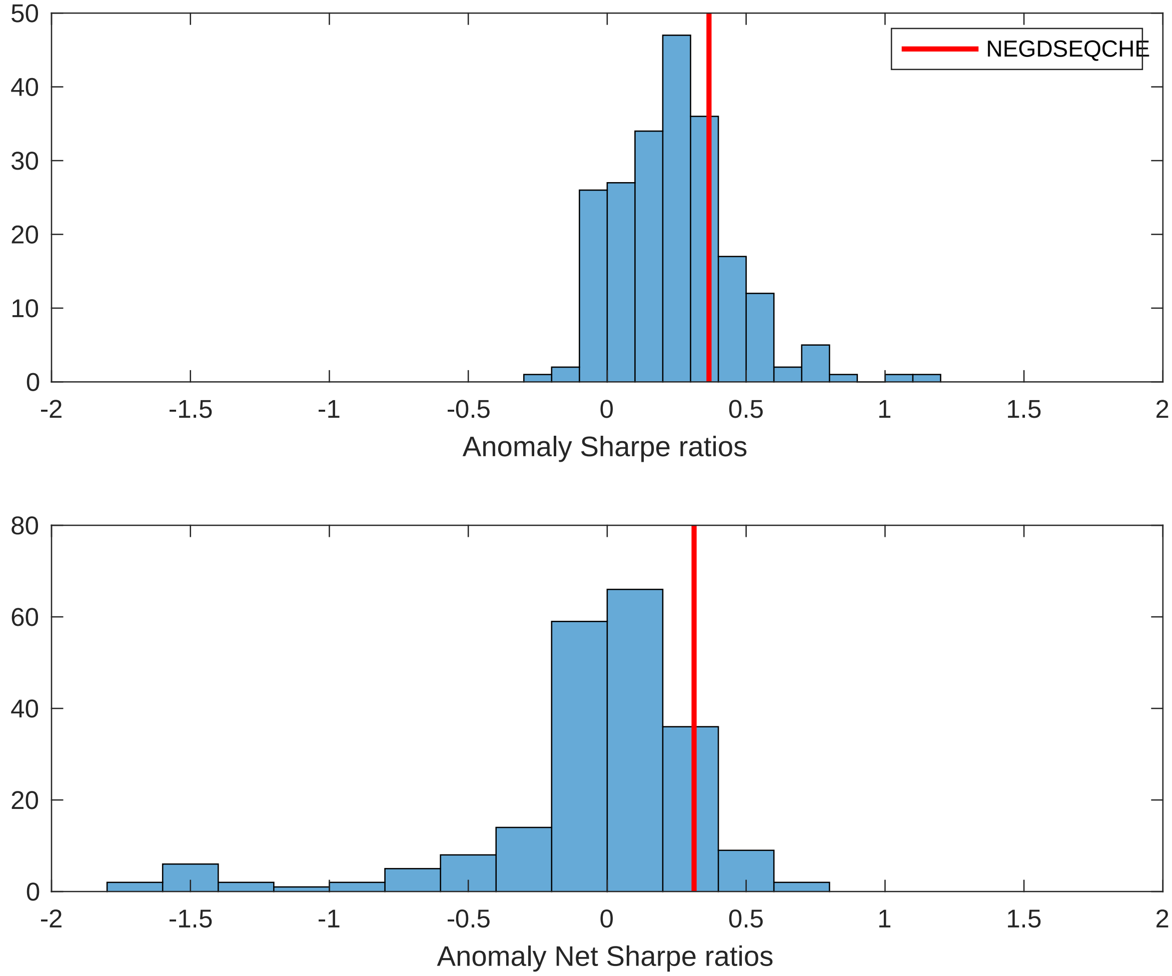


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

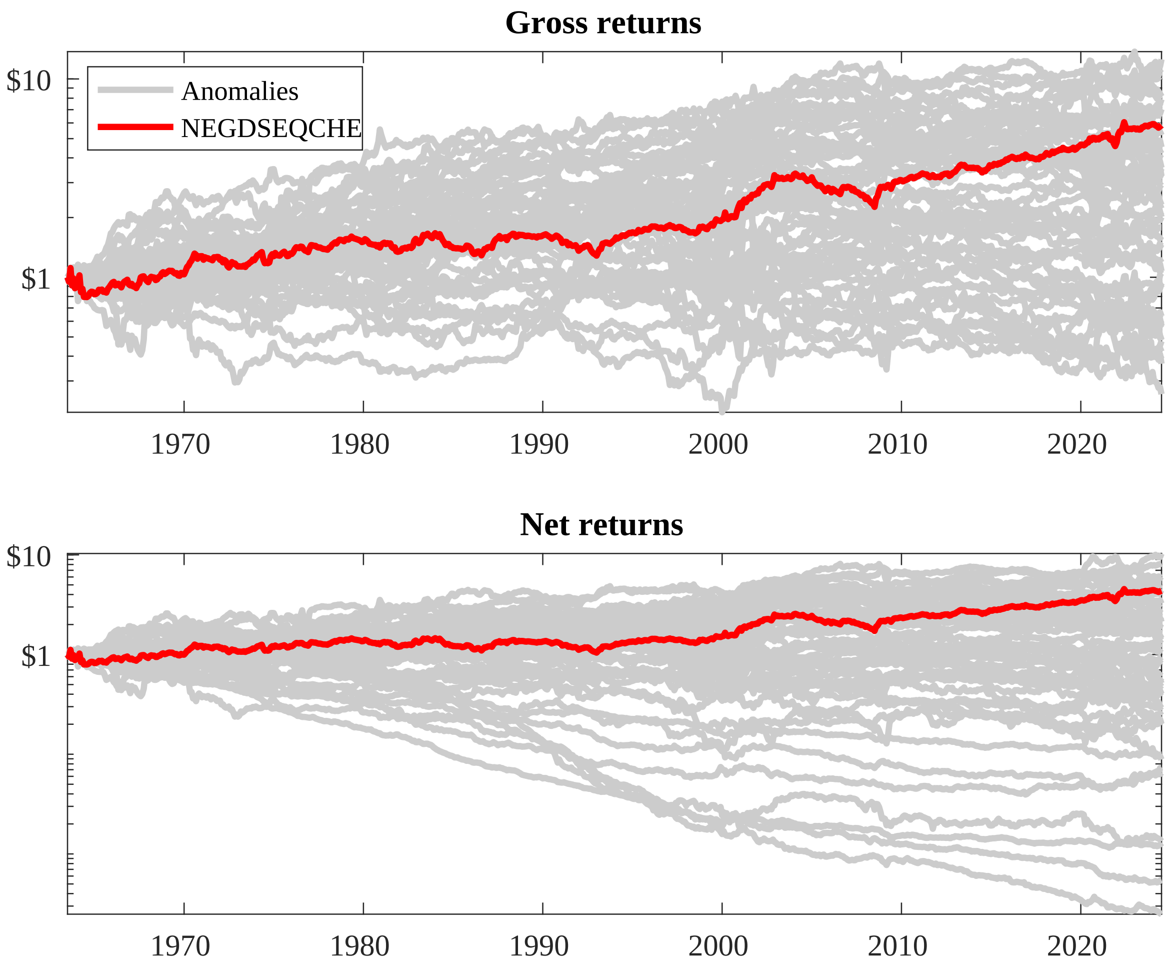
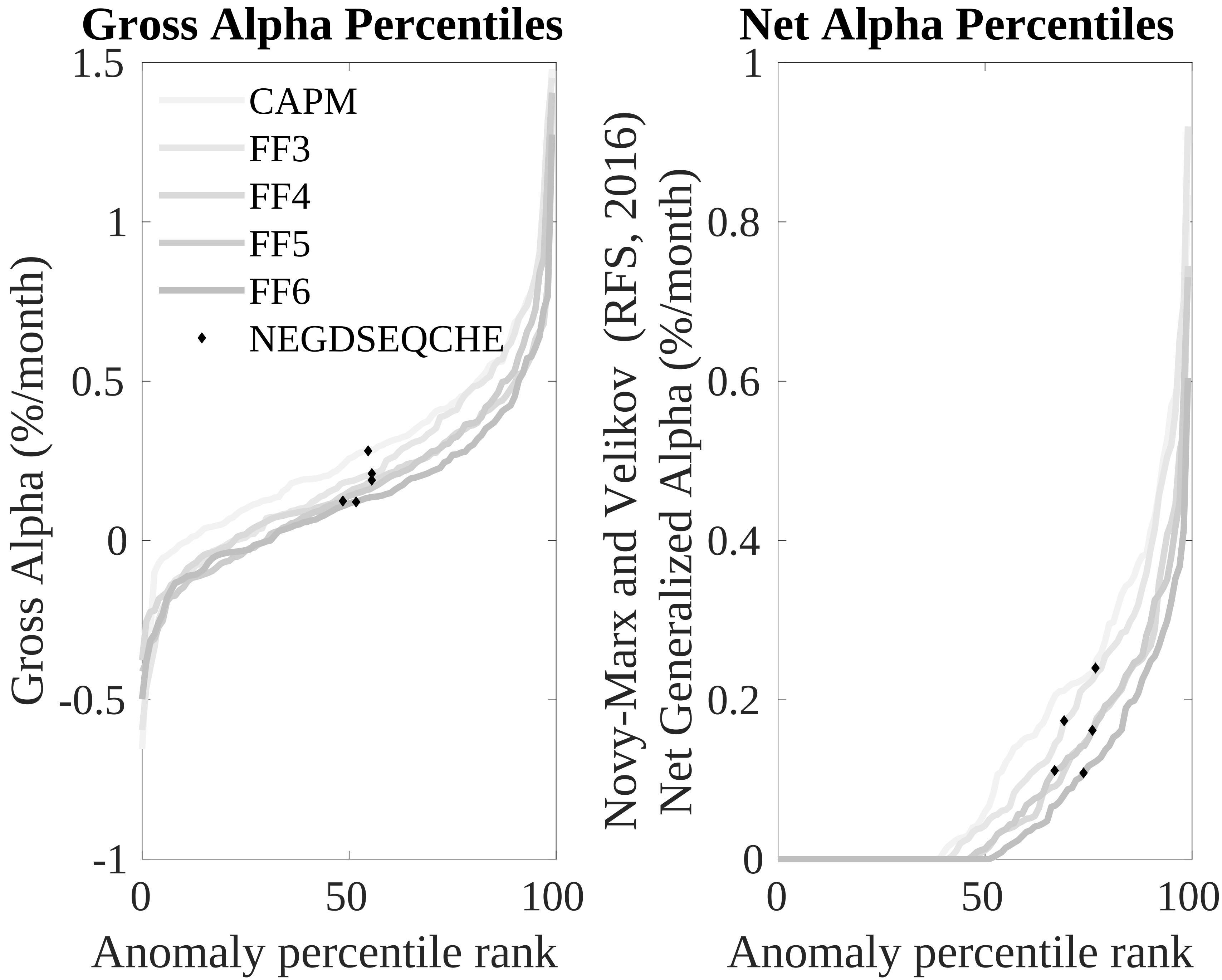


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDSEQCHE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

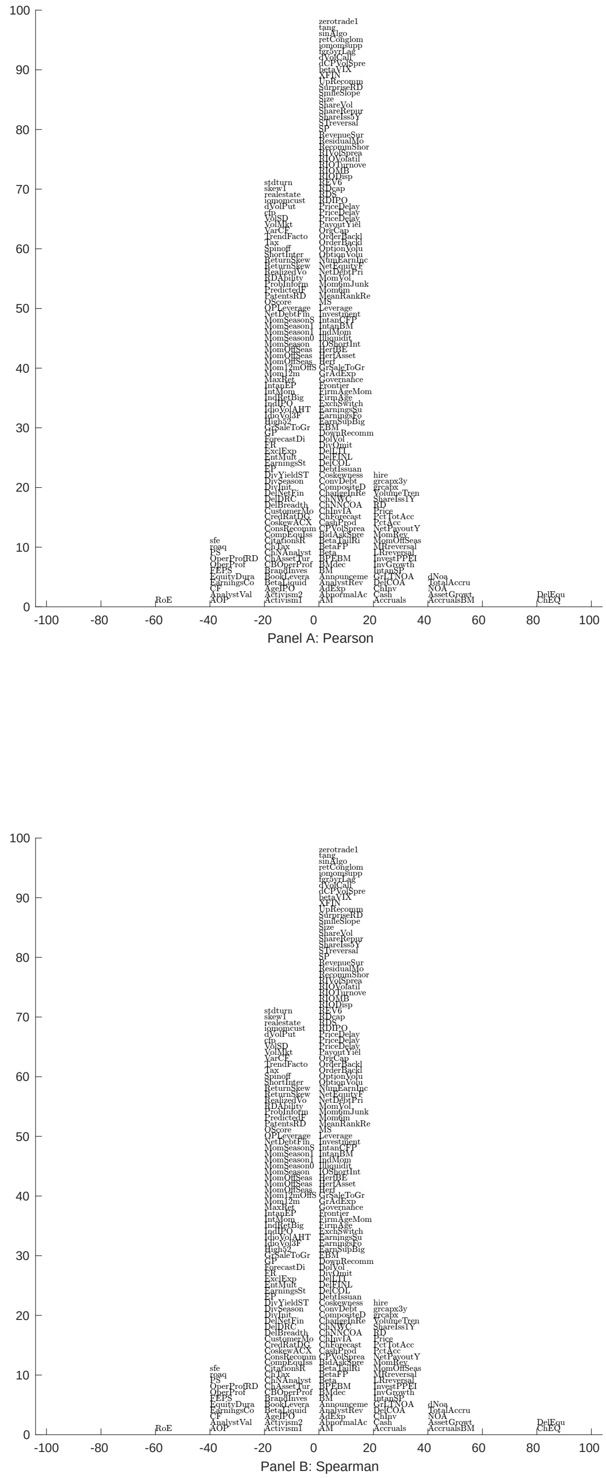


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

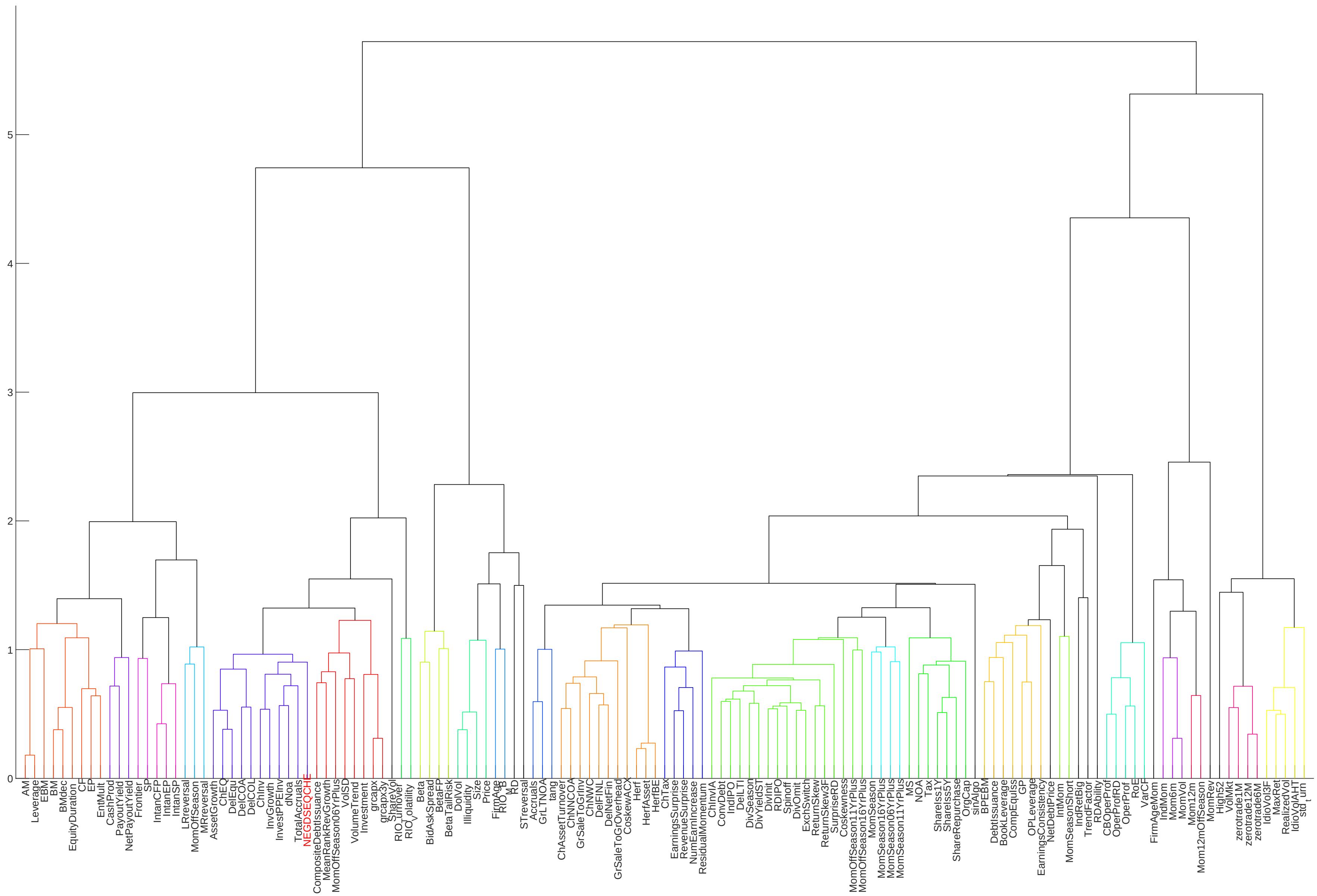


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

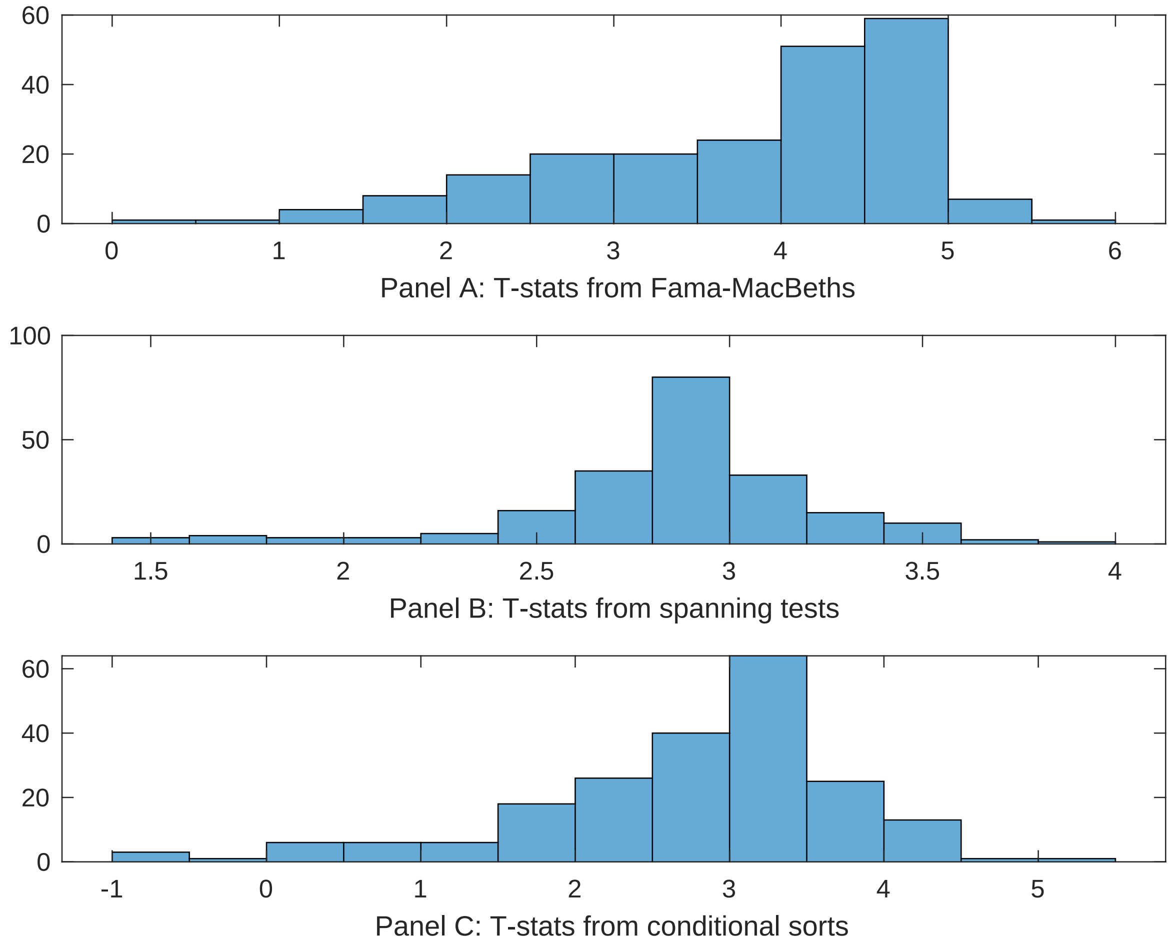


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDI. Stocks are finally grouped into five EDI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDI} EDI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.17	0.13	0.13	0.12	0.11	0.12
	[5.67]	[7.39]	[6.10]	[5.94]	[5.40]	[3.92]	[5.05]
EDI	0.14	0.16	0.10	0.13	0.24	0.12	0.36
	[3.15]	[3.60]	[2.20]	[2.52]	[4.43]	[3.34]	[1.15]
Anomaly 1	0.14						0.62
	[4.00]						[1.50]
Anomaly 2		0.44					0.12
		[4.31]					[0.13]
Anomaly 3			0.99				0.48
			[9.29]				[3.73]
Anomaly 4				0.13			0.41
				[9.91]			[2.36]
Anomaly 5					0.34		-0.25
					[1.82]		[-1.02]
Anomaly 6						0.13	0.59
						[3.11]	[1.37]
# months	708	708	708	708	708	432	432
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Asset growth, change in net operating assets, Total accruals, Percent Total Accruals. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.10	0.12	0.08	0.09	0.26	0.23
	[1.48]	[1.15]	[1.36]	[0.87]	[1.03]	[2.87]	[2.82]
Anomaly 1	41.89						10.17
	[9.60]						[1.24]
Anomaly 2		47.92					47.47
		[10.11]					[6.50]
Anomaly 3			23.29				-21.05
			[4.24]				[-3.59]
Anomaly 4				29.57			13.79
				[5.77]			[2.67]
Anomaly 5					20.50		-1.40
					[4.85]		[-0.27]
Anomaly 6						19.94	11.17
						[4.95]	[2.70]
mkt	5.67	8.01	6.47	6.58	5.73	-0.55	1.25
	[2.70]	[3.84]	[2.94]	[3.02]	[2.61]	[-0.24]	[0.61]
smb	5.49	4.42	4.16	6.34	6.11	4.84	2.99
	[1.81]	[1.46]	[1.30]	[2.02]	[1.93]	[1.48]	[1.01]
hml	-14.74	-15.02	-9.91	-11.46	-8.96	-5.97	-9.73
	[-3.60]	[-3.69]	[-2.33]	[-2.71]	[-2.12]	[-1.49]	[-2.61]
rmw	-5.61	-7.19	-9.78	-9.20	-6.21	-13.58	-17.18
	[-1.37]	[-1.77]	[-2.28]	[-2.17]	[-1.44]	[-3.25]	[-4.52]
cma	20.34	16.08	34.42	40.47	52.23	51.70	15.36
	[2.75]	[2.15]	[3.78]	[5.57]	[7.95]	[8.67]	[1.99]
umd	1.46	-0.38	1.21	-0.13	0.98	0.61	-1.85
	[0.70]	[-0.18]	[0.55]	[-0.06]	[0.45]	[0.30]	[-0.99]
# months	731	731	731	731	731	438	438
$\bar{R}^2$(%)	27	28	19	21	20	31	46

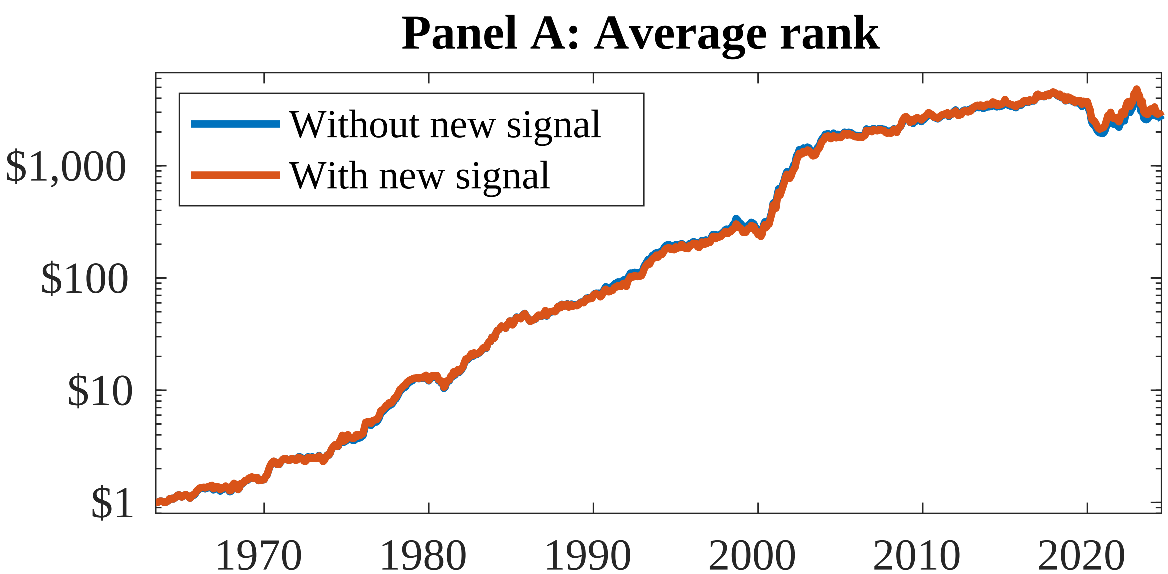


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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